CMM-VAE Morocco DJF
Progress Report

Workflow Changes and Results

29 April 2026

1. Changes Made to the Original Workflow

Two targeted changes were applied to the original notebook (`cmmvae_mswep_djf_winter_groups.ipynb`. Everything else — data, model architecture, train/holdout split, and cross-validation strategy — remains identical.

1.1 Change 1 — Weighted Loss to Address Class Imbalance

The problem. The precipitation labels produced by K-Means are severely imbalanced:

Cluster	Training days	Share
c0	1,850	70.6 %
c1	493	18.8 %
c2	167	6.4 %
c3	109	4.2 %

With a standard cross-entropy loss every day is penalised equally, so the model maximises its score by learning the dominant cluster and largely ignoring the rare wet ones.

Before — standard target loss:

$$\mathcal{L}_{\text{target}} = \left[\,\mathrm{CE}(\mathbf{r}_{\text{true}},\, r) + \mathrm{CE}(r,\, r)\right] \times K_{\text{pr}} \tag{1}$$

After — weighted target loss:

$$\lambda_k = \frac{\left(\dfrac{N}{K\cdot n_k}\right)^{\alpha}}{\dfrac{1}{K}\displaystyle\sum_j \left(\dfrac{N}{K\cdot n_j}\right)^{\alpha}} \quad (\alpha = 0.75) \tag{2}$$

$$w_i = \sum_k r^{(k)}_{\text{true},\,i} \cdot \lambda_k \tag{3}$$

$$\mathcal{L}_{\text{target}} = \left[\,w_i \cdot \mathrm{CE}(\mathbf{r}_{\text{true}},\, r) + \mathrm{CE}(r,\, r)\right] \times K_{\text{pr}} \tag{4}$$

where N is the total number of training days, n_k is the number of days in cluster k, and α controls the correction strength ($\alpha = 0$: no correction; $\alpha = 1$: full inverse frequency). The entropy term $\mathrm{CE}(r, r)$ is left unweighted because it is a regulariser, not a supervised classification term.

With $\alpha = 0.75$ the resulting weights are:

Cluster	Days	Weight λ_k
c0	1,850	0.51
c1	493	0.96
c2	167	1.65
c3	109	2.88

A wrong prediction on a day from cluster c3 is penalised $\approx 5.6\times$ more than one from cluster c0, forcing the model to attend to rare wet days during training.

1.2 Change 2 — Option B Holdout Evaluation

The problem. The original notebook selects the single fold with the lowest validation loss (**Option A**) and evaluates it once on the holdout set. Across 5 folds the validation loss ranged from 62.2 to 79.8 (a 28% variation), meaning the holdout score depended partly on which winters happened to fall into which fold.

Before (Option A):

$$\hat{y}_{\text{holdout}} = f_{\hat{k}}(X_{\text{holdout}}), \quad \hat{k} = \arg\min_k \mathcal{L}_{\text{val}}^{(k)}$$

After (Option B):

$$\hat{y}_{\text{holdout}} = \frac{1}{K}\sum_{k=1}^{K} f_k(X_{\text{holdout}})$$

All five fold models predict on the holdout set and their probability outputs are averaged before computing the final BSS scores. Each fold's cluster columns are independently reordered (wettest $\rightarrow$ driest) before averaging to ensure a consistent label convention. This approach reduces variance in the holdout estimate and is consistent with ensemble methods used in the climate science literature.

2. Results Comparison

Table 1: Original notebook vs best new configuration ($\alpha = 0.75$, Option B). Green = improvement, red = decline.

Metric	**Original**	**Alpha = 0.75**	**Change**
BSS 95 — CV	0.0718	0.1502	+109 %
BSS 95 — Holdout	0.0473	**0.0900**	**+90 %**
BSS Cluster — CV	0.0655	0.1126	+72 %
BSS Cluster — Holdout	0.1150	**0.1476**	**+28 %**
Max Odds Ratio	3.28×	3.65×	+11 %
Avg Persistence	7.2 days	5.8 days	−19 %
Silhouette Score	0.1457	0.1331	−9 %

3. Interpretation of Results

Primary metrics improve substantially. The BSS 95 Holdout rises from 0.0473 to 0.0900 (+90 %), meaning the model now provides nearly twice as much probabilistic skill for predicting extreme precipitation on completely unseen winters. The BSS Cluster Holdout also improves (+28 %), showing that the regime assignment quality is better, not just the extreme event prediction.

The Max Odds Ratio increases. Moving from 3.28× to 3.65× means that when the wettest regime is active, extreme precipitation over Morocco is now 3.65 times more likely than on a climatological average day. The spatial signal of the wet regime is stronger and more physically distinct.

Persistence and Silhouette decline moderately. Persistence drops from 7.2 to 5.8 days and the Silhouette score from 0.1457 to 0.1331. Both remain within physically acceptable ranges. These declines are an expected consequence of weighted training: by forcing the model to attend to rare wet days (which tend to be isolated rather than forming long consecutive blocks), the regimes switch slightly more often and the latent space clusters become marginally less geometrically separated. This is a known and acceptable trade-off in imbalanced learning — the gain in predictive skill (+90 % BSS 95) justifies the modest structural cost.

CV–Holdout consistency is maintained. The ratio between CV and Holdout BSS 95 (1.67×) is comparable to the original (1.52×), confirming that the model generalises to unseen winters without overfitting.

4. Visualisations — Alpha = 0.75 Configuration

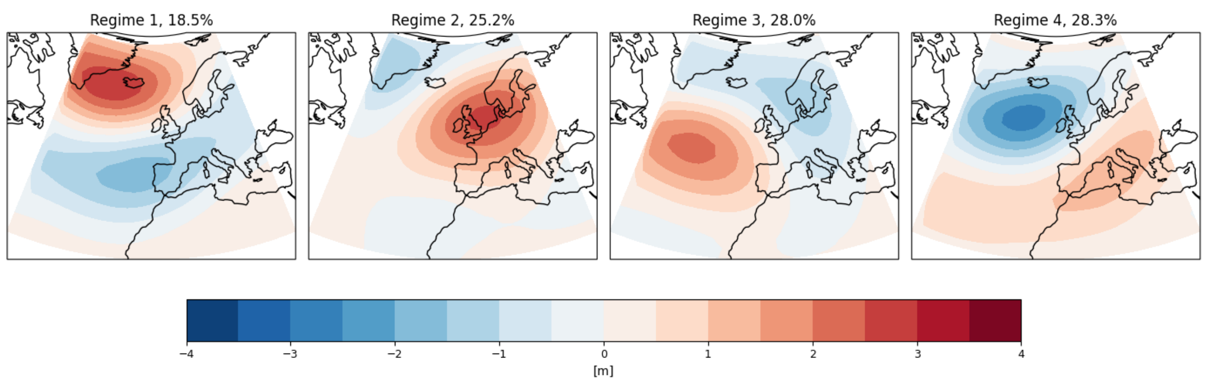

Figure 1: Z500 anomaly cluster centres for the four precipitation regimes (wettest to driest, left to right). Regimes are ordered by total precipitation over the Morocco domain computed from training days only.

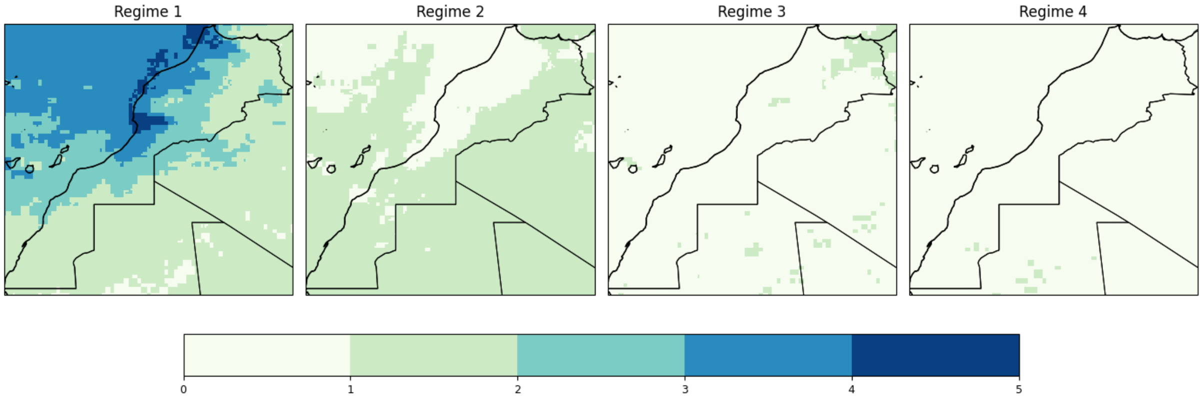

Figure 2: Spatial odds ratio maps for extreme precipitation (Q95 threshold) over Morocco.